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Principles of Macroeconomics

Lecture 9: Business cycles

Prof. Dr. Jan-Egbert Sturm

Schedule

Date	Lecture	Topic	Chapters
16/09/2025	1	Introduction	1, 2
23/09/2025	2	National Accounting	20
30/09/2025	3	Production and Growth	21
07/10/2025	4	Unemployment and the labour market	22
14/10/2025	5	Saving, investment and the financial system	23
21/10/2025	6	The Monetary System	24
28/10/2025	7	International Trade	3,15,17
04/11/2025	8	Open-Economy Macroeconomics	25
11/11/2025	9	Business Cycles	26
18/11/2025	10	Keynesian Economics and IS-LM Analysis	27
25/11/2025	11	Aggregate Demand and Aggregate Supply	28
02/12/2025	12	Monetary and Fiscal Policy	29
09/12/2025	13	Phillips Curve / Supply-side policies	30,31
16/12/2025	14	Exam preparation	

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Principles of Macroeconomics

Lecture 8: Open-economy macroeconomics

Prof. Dr. Jan-Egbert Sturm

Real Exchange Rates

- The ***real exchange*** rate is the rate at which a person can trade the goods and services of one country for the goods and services of another
- The real exchange rate compares the prices of domestic goods and foreign goods in the domestic economy
  - If a case of Swiss beer is twice as expensive as German beer, the real exchange rate is 2 cases of German beer per case of Swiss beer.
- The real exchange rate depends on the nominal exchange rate (foreign currency / local currency = fc/lc) and the prices of goods in the two countries measured in local currencies
- The real exchange rate is a key determinant of how much a country exports and imports

$$\textit{Real exchange rate} = \frac{\textit{Nominal exchange rate } (fc/lc) \times \textit{Domestic price}}{\textit{Foreign price}}$$

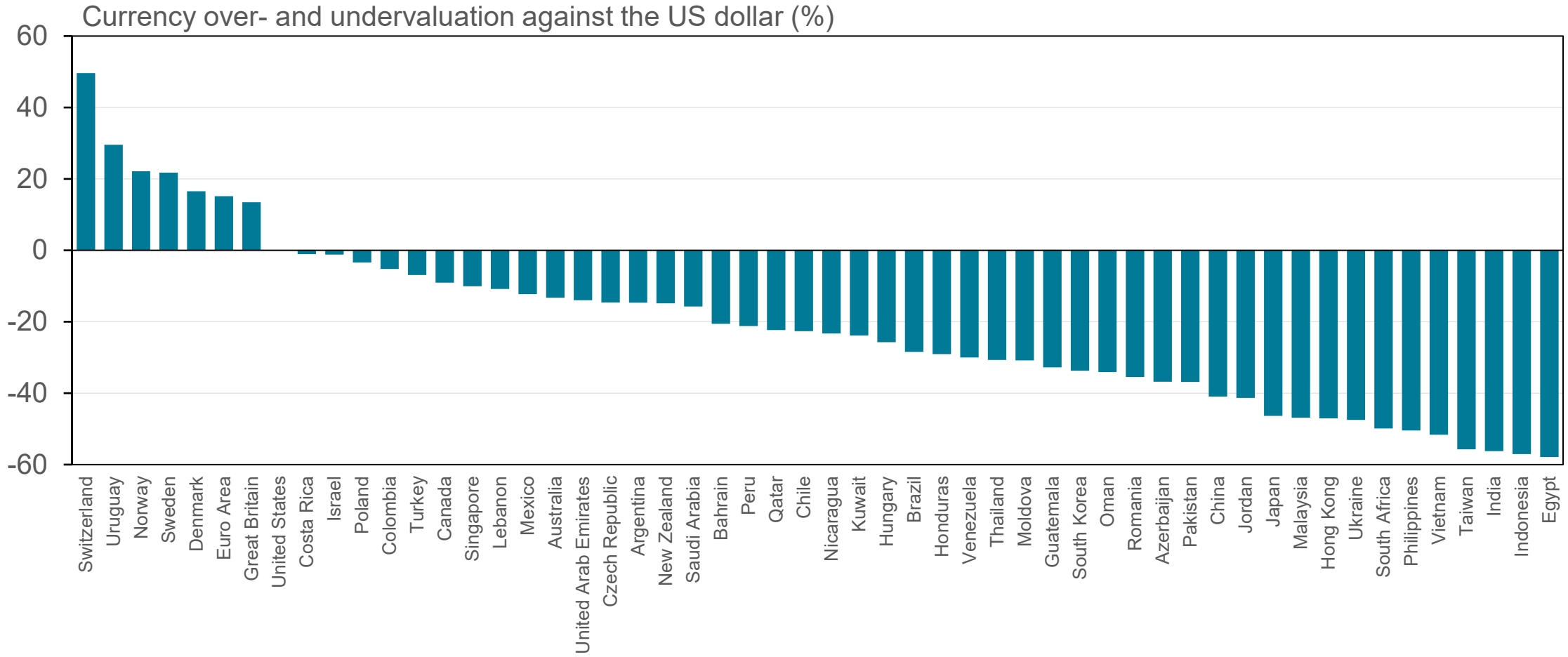
McZample

- one good: Big Mac
- price in the United States $P^* = \$\ 6.01$
- price in Switzerland: $P = \text{CHF}\ 7.20$
- nominal exchange rate: $e = 1.25\ \$/\text{CHF}$

- Real exchange rate: $\varepsilon = 1.25 * 7.20 / 6.01 = 1.50$
  - (Overvaluation of the Swiss France relative to the US Dollar: 50%)

To buy a Swiss Big Mac, someone from the US would have to pay an amount that could buy 1.5 Big Macs in the US

The Big-Mac-Index



Source: The Economist (July 2025)

The Big Mac Index (The Economist, July 2025)

- In America, a McDonald's Big Mac costs $6.01
- In Taiwan, the same burger costs just NT$ 78, equivalent to $2.66 at market exchange rates
  - Since NT$ 78 can buy as much burger as US$ 6.01, a US$ should be worth NT$ 12.98
  - In fact, a US$ costs NT$ 19.31, suggesting that the NT$ is undervalued by about 56%
- The euro was overvalued by 15% relative to the US dollar
- The Swiss franc is the most expensive currency (for buying a big mac)
  - It was overvalued by 50% relative to the US dollar in July 2025

Source: https://www.economist.com/big-mac-index

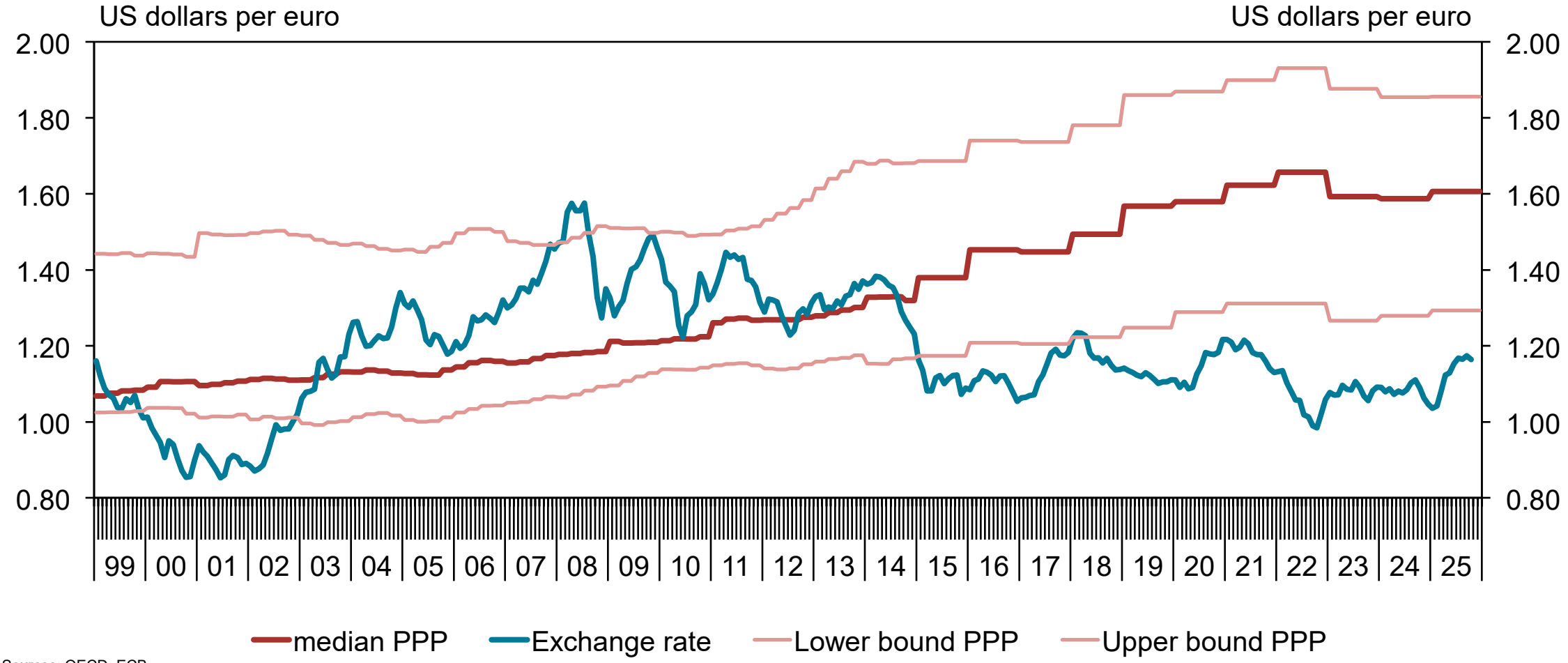
Real Exchange Rates

- A depreciation (fall) in the Swiss real exchange rate means that Swiss goods have become cheaper relative to foreign goods
- This encourages consumers both at home and abroad to buy more Swiss goods and fewer goods from other countries
- As a result, Swiss exports rise, and Swiss imports fall, and both of these changes raise Swiss net exports
- Conversely, an appreciation in the Swiss real exchange rate means that Swiss goods have become more expensive compared to foreign goods, so Swiss net exports fall

A First Theory of Exchange-Rate Determination: Purchasing-Power Parity

- The ***purchasing-power parity (PPP) theory*** is the simplest and most widely accepted theory explaining the variation of currency exchange rates
  - PPP is a theory of exchange rates whereby a unit of any given currency should be able to buy the same quantity of goods in all countries
- The PPP theory is based on a principle called the law of one price
  - According to the law of one price, a good must sell for the same price in all locations
  - If the law of one price were not true, unexploited profit opportunities would exist
    - The process of taking advantage of differences in prices in different markets is called arbitrage
      - If arbitrage occurs, eventually prices that differed in two markets would necessarily converge
- According to the PPP theory, a currency must have the same purchasing power in all countries and exchange rates move to ensure that
  - The nominal exchange rate between two currencies reflects the different price levels in the countries
- Limitations of the PPP theory
  - Many goods are not easily traded or shipped from one country to another
  - Tradable goods are not always perfect substitutes when they are produced in different countries

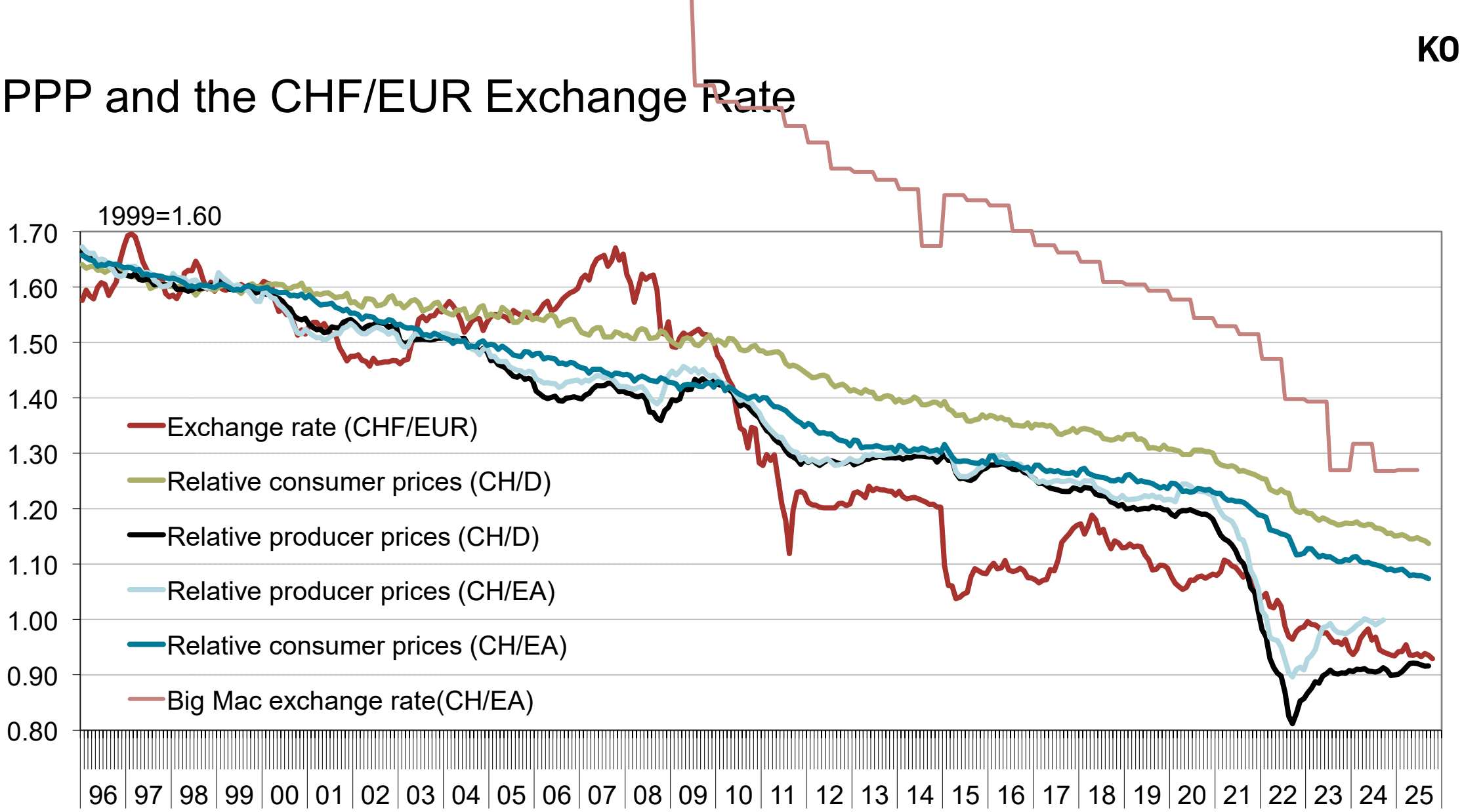
Exchange rate of the euro against the US dollar and PPP



Sources: OECD, ECB

Notes: The nominal exchange rate is based on monthly data, while the exchange rate based on purchasing power parity (PPP) is given at a quarterly frequency. The US dollar-euro PPP rate is calculated as the GDP-weighted average of the euro country-specific PPP estimates vis-à-vis the US dollar. The PPP upper bound represents the 95th percentile of the euro country-specific PPP estimates vis-à-vis the US dollar; the lower bound the 5th percentile. In calculating these bounds, the 11 euro area member countries with the largest GDP weights are used.

PPP and the CHF/EUR Exchange Rate



Sources: Eurostat, OECD, Macrobond

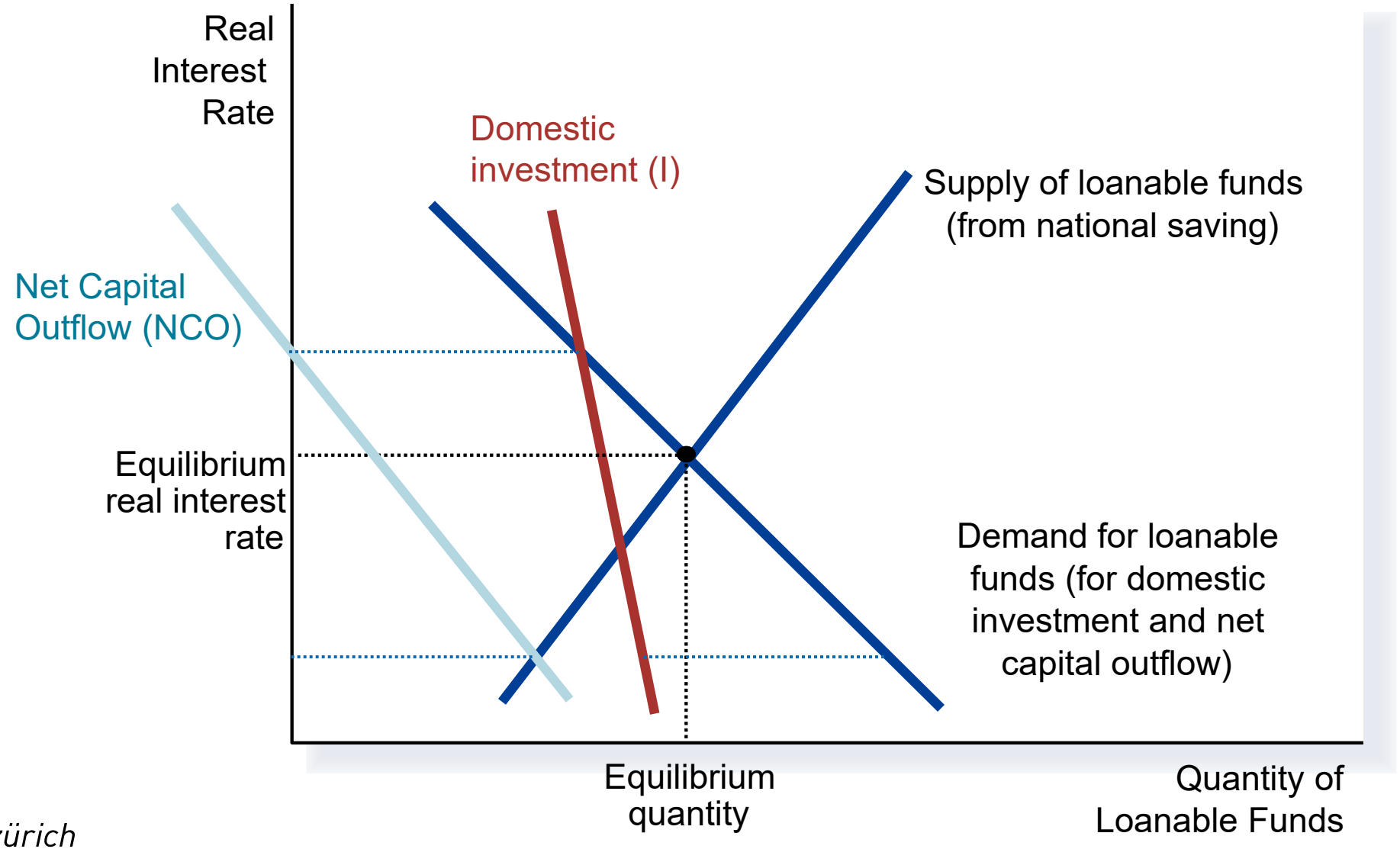
A Macroeconomic Theory of the Open Economy

- Key Macroeconomic Variables in an Open Economy
  - The important macroeconomic variables of an open economy include:
    - net exports
    - net foreign investment (= net capital outflow)
    - nominal exchange rates
    - real exchange rates
- Basic Assumptions of a Macroeconomic Model of an Open Economy
  - The model takes the economy's GDP as given
  - The model takes the economy's price level as given (nominal=real)
- To understand forces at work in an open economy, we must consider two markets:
  - The market for loanable funds
    - Co-ordinates savings, investment and net capital outflow
  - The market for foreign currency
    - Co-ordinates people who wish to exchange domestic to foreign currency

The Market for Loanable Funds

- The Market for Loanable Funds: $S = I + NCO$
- The supply of loanable funds comes from national saving (*S*)
  - A higher real interest rate encourages people to save, i.e. increase the supply of loanable funds
- The demand for loanable funds comes from domestic investment (*I*) and net capital outflows (*NCO*)
  - A higher real interest rate discourages firms to invest in new buildings, machines & equipment
  - A higher real interest rate discourages people to buy assets abroad
- The interest rate adjusts to bring supply and demand for loanable funds into balance
  - At the equilibrium interest rate, the amount that people want to save exactly balances the desired quantities of investment and net capital outflows

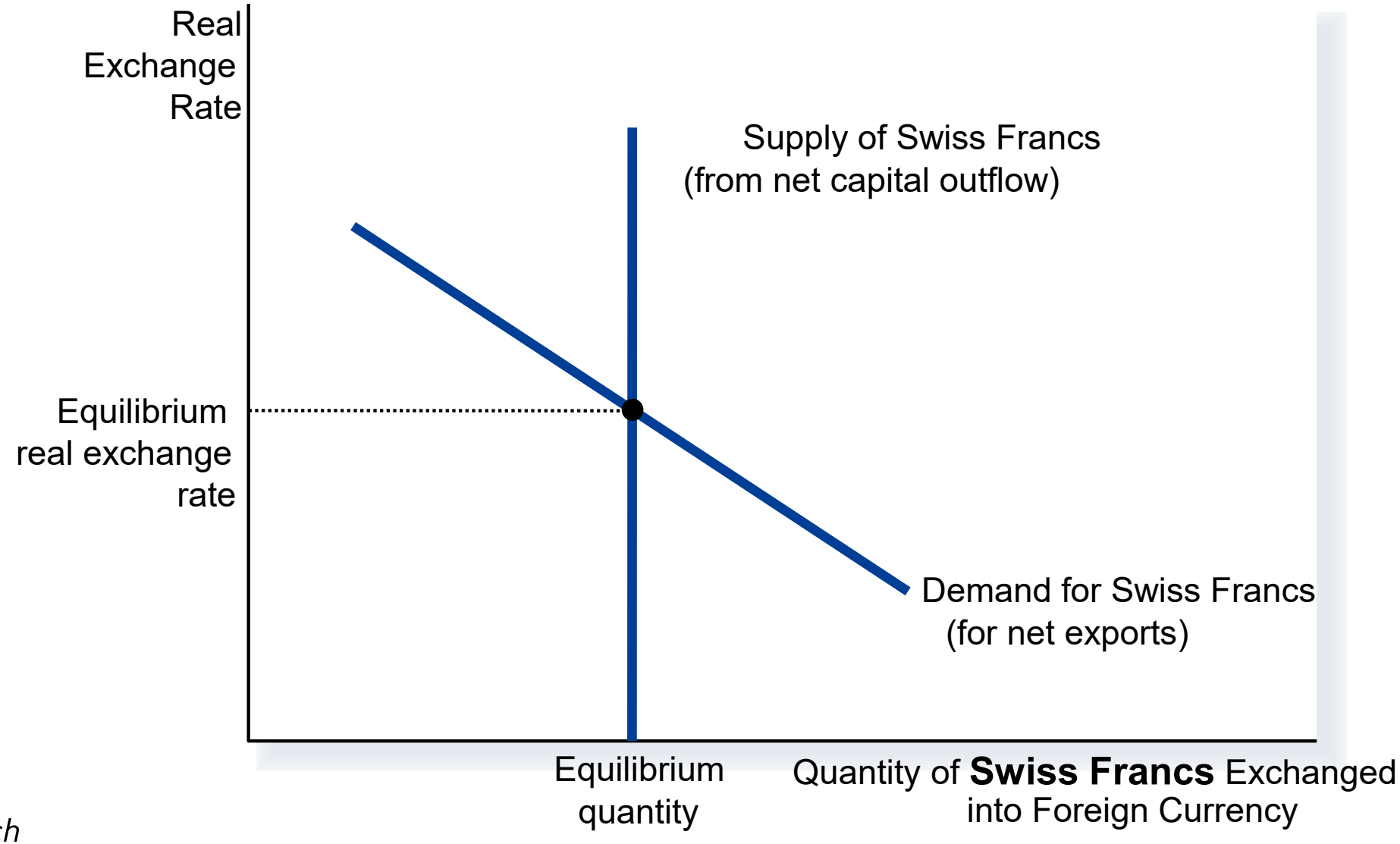
The Market for Loanable Funds



The Market for Foreign-Currency Exchange

- The two sides of the foreign currency exchange market are represented by NCO and NX
  - NCO (net capital outflow) represents the imbalance between the purchases and sales of assets
  - NX (net exports) represents the imbalance between exports and imports of goods and services
- For an economy as a whole, NCO and NX must balance, or: ***NCO = NX***
  - Supply of domestic currency = demand for domestic currency
- In the market for foreign-currency exchange, Swiss Francs are traded for foreign currencies
  - The price that balances the supply and demand for foreign-currency is the real exchange rate
  - A real appreciation of the Swiss Franc will be defined as a higher level of its real exchange rate
- A low real exchange rate of the Swiss Franc, leads to strong exports and a high demand for its currency
  - The demand curve (*NX*) for domestic currency is downward sloping
  - The supply curve (*NCO*) is vertical because the quantity of Swiss Francs supplied is unrelated to the (current) real exchange rate
    - It does depend upon the expected change in the real exchange rate during the investment horizon
    - We (implicitly) assume that the best forecast for the future real exchange rate is its current level
      - This implies that the (expected) change in the real exchange rate equals zero

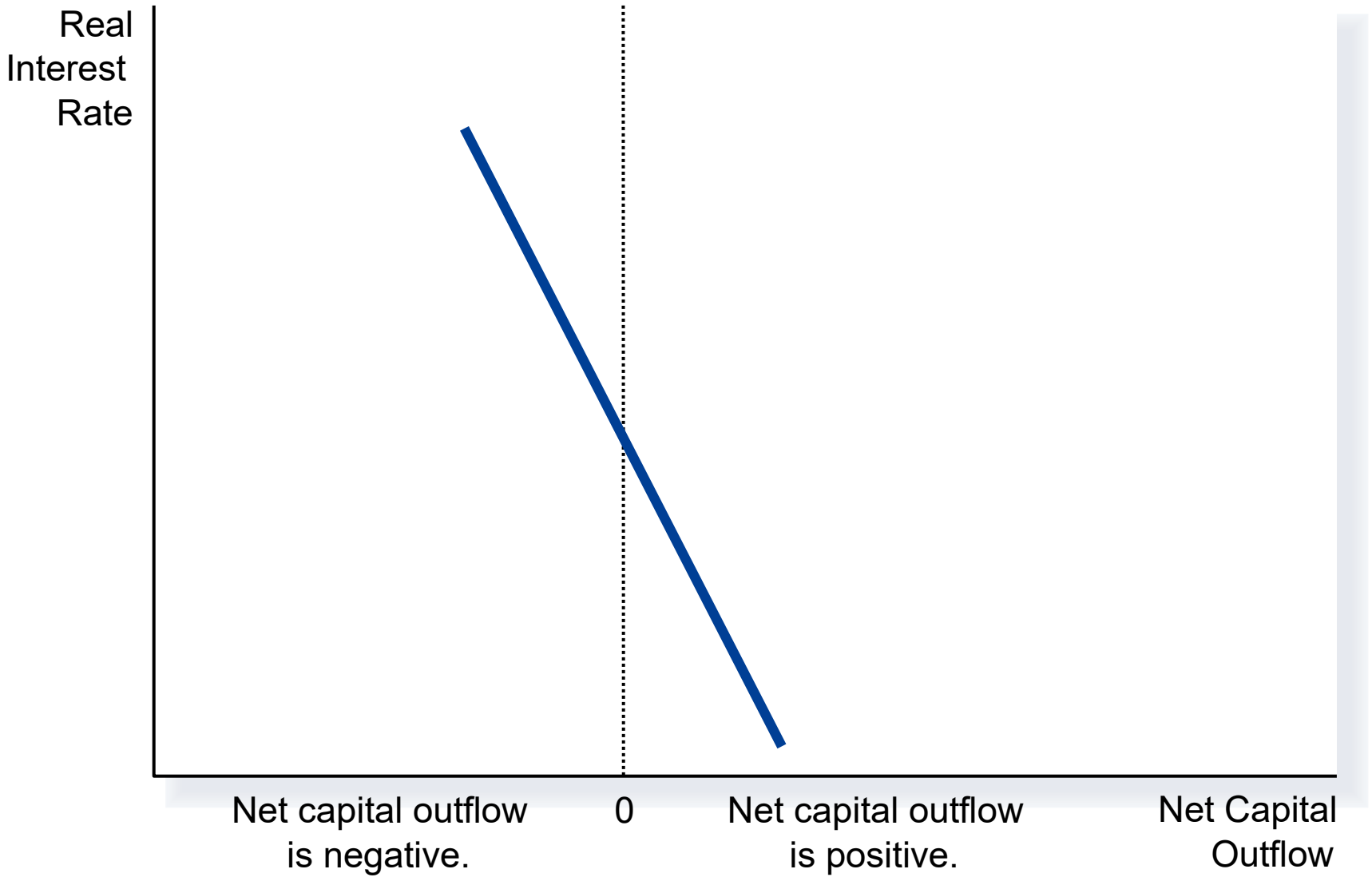
The Market for Foreign-Currency Exchange



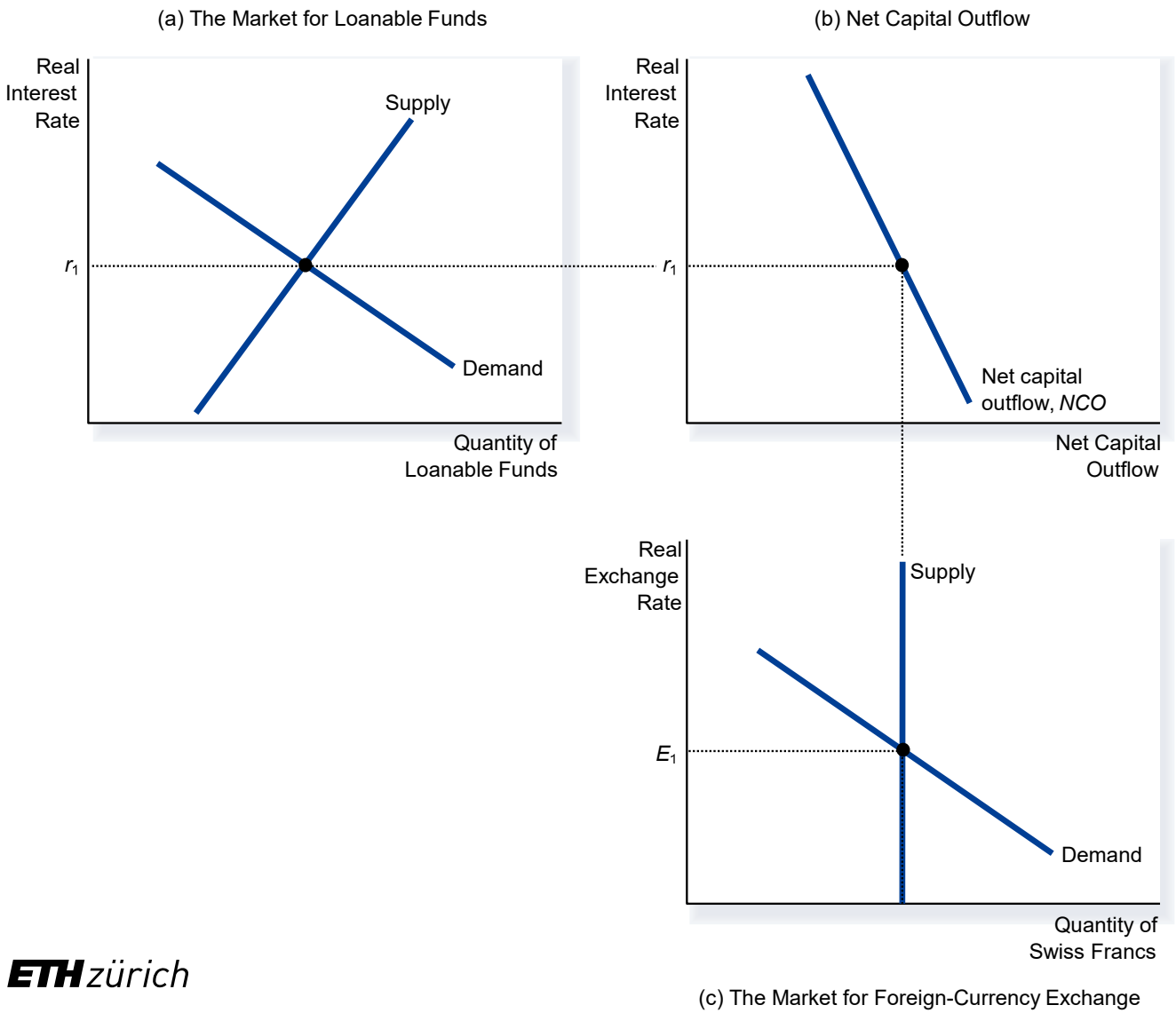
Equilibrium in the Open Macroeconomy

- The Market for Foreign-Currency Exchange
  - The real exchange rate adjusts to balance the supply and demand for Swiss Francs
    - At the equilibrium real exchange rate, the demand for Swiss Francs to buy net exports exactly balances the supply of Swiss Francs to be exchanged into foreign currency to buy assets abroad
- In the market for ***loanable funds***, supply comes from national saving and demand comes from domestic investment and net capital outflow.
- In the market for ***foreign currency exchange***, supply comes from net capital outflow and demand comes from net exports
- Equilibrium in the Open Economy
  - The market for loanable funds and the market for foreign exchange are related to each other
  - Net capital outflow links the loanable funds market and the foreign currency exchange market
  - The key determinant of net capital outflow is the real interest rate

How Net Capital Outflow Depends on the Interest Rate



Simultaneous Equilibrium in the Two Markets

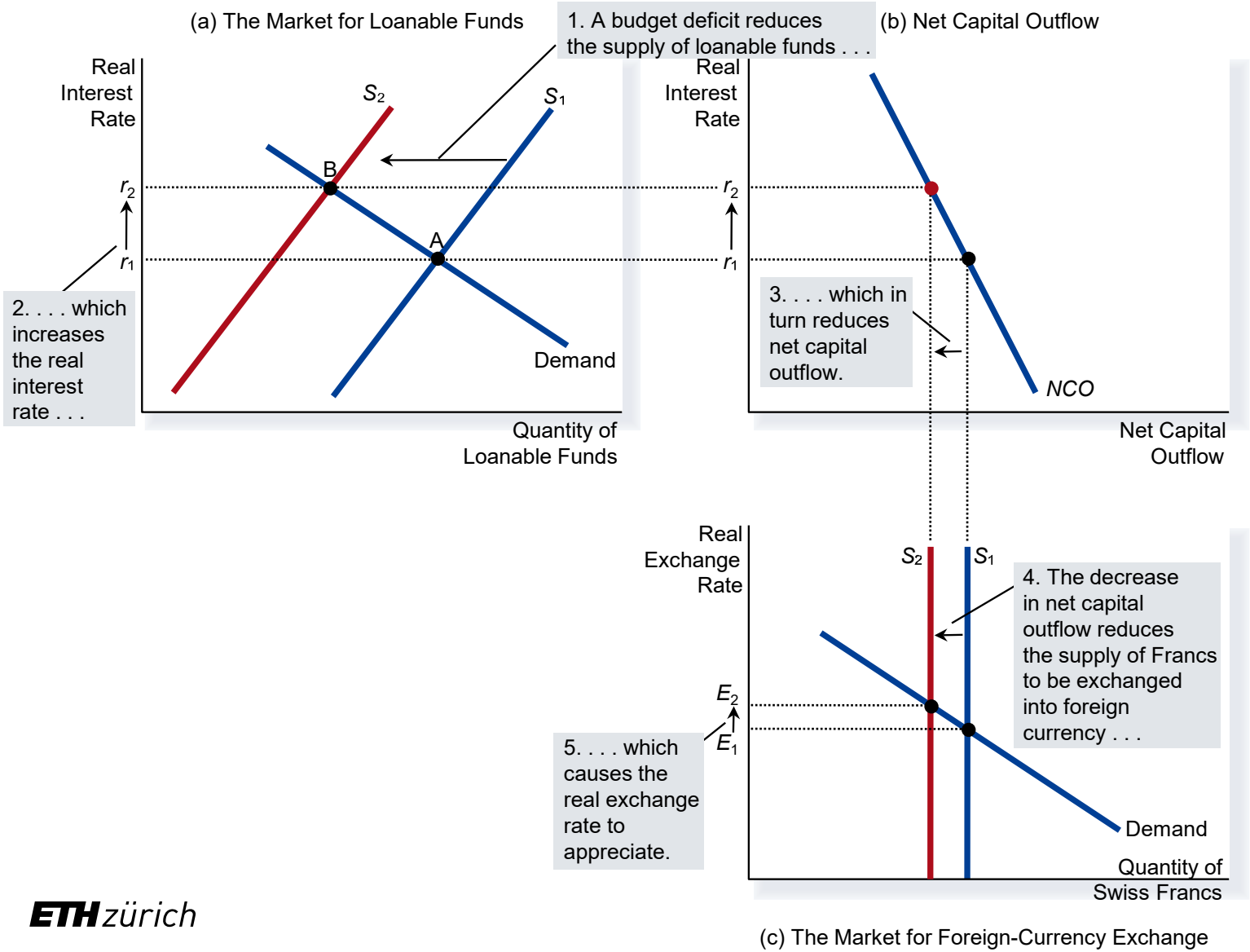


- Prices in the loanable funds market and the foreign-currency exchange market adjust simultaneously to balance supply and demand in these two markets
- As they do, they determine the macroeconomic variables of national saving, domestic investment, net foreign investment (net capital outflow), and net exports

How Policies and Events Affect an Open Economy

- The magnitude and variation in important macroeconomic variables depend on the following:
  - Government budget deficits
  - Trade policies
  - Political and economic stability

The Effects of Government Budget Deficit



- In an open economy, government budget deficits
  - reduce the supply of loanable funds,
  - drive up the interest rate,
  - crowd out domestic investment,
  - cause net foreign investment (net capital outflow) to fall
  - reduces the supply of Swiss Francs to be exchanged into foreign currency
  - Causes the real exchange rate to appreciate

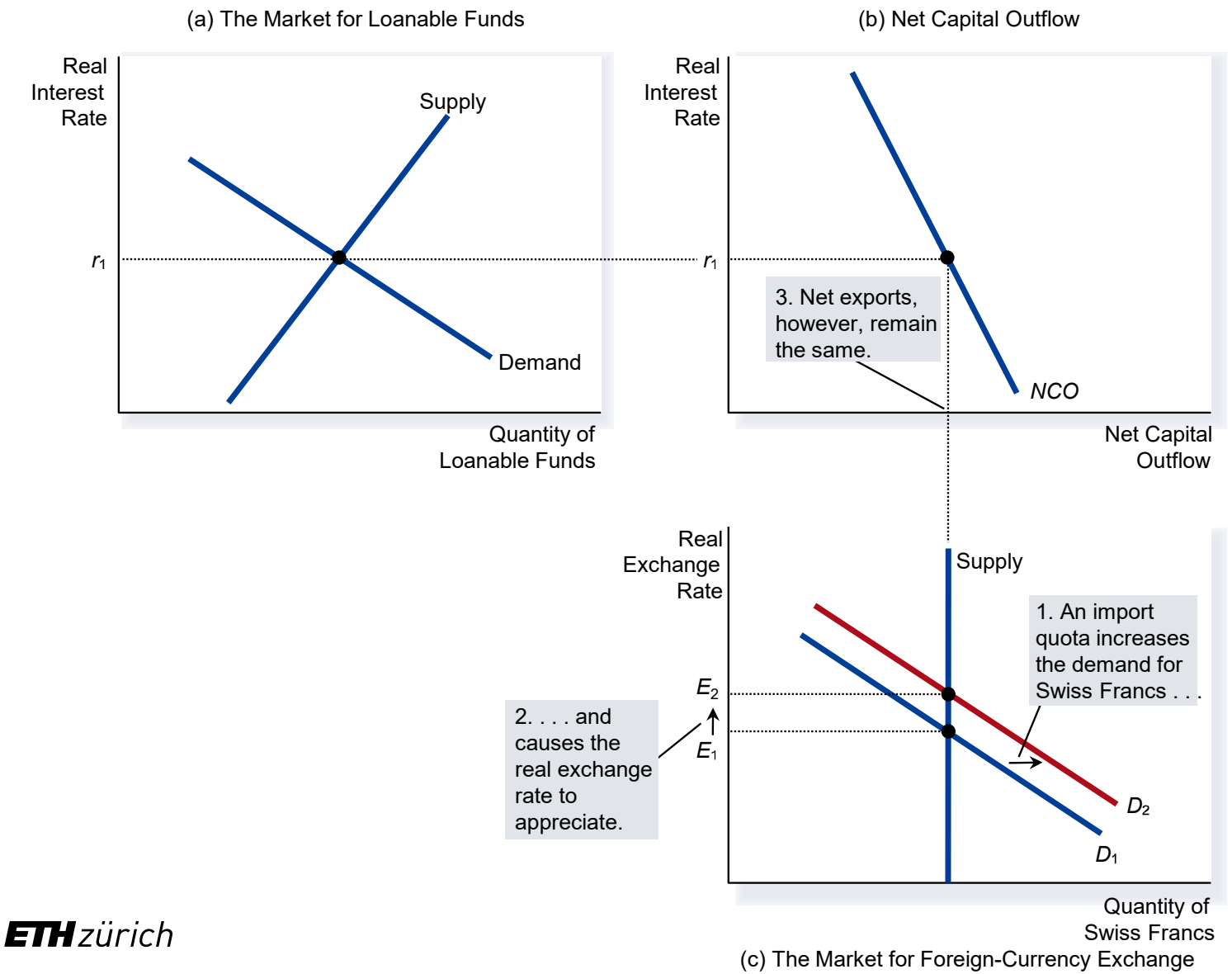
Trade Policy

- A trade policy is a government policy that directly influences the quantity of goods and services that a country imports or exports.
  - Tariff: A tax on an imported good
  - Import quota: A limit on the quantity of a good produced abroad and sold domestically
- Because they do not change national saving or domestic investment, trade policies do not affect the trade balance
  - For a given level of national saving and domestic investment, the real exchange rate adjusts to keep the trade balance the same
  - Trade policies have a greater effect on microeconomic than on macroeconomic markets

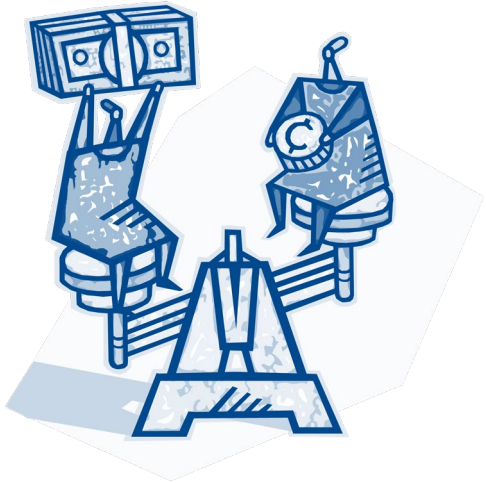
Trade Policy: Effect of an Import Quota

- Because foreigners need Swiss Francs to buy Swiss net exports, there is an increased demand for Swiss Francs in the market for foreign-currency
  - This leads to an appreciation of the real exchange rate
- There is no change in the interest rate because nothing happens in the loanable funds market
- There will be no change in net exports
- There is no change in net foreign investment even though an import quota reduces imports
- An appreciation of the Swiss Franc in the foreign exchange market encourages imports and discourages exports
- This offsets the initial increase in net exports due to the import quota

The Effects of an Import Quota



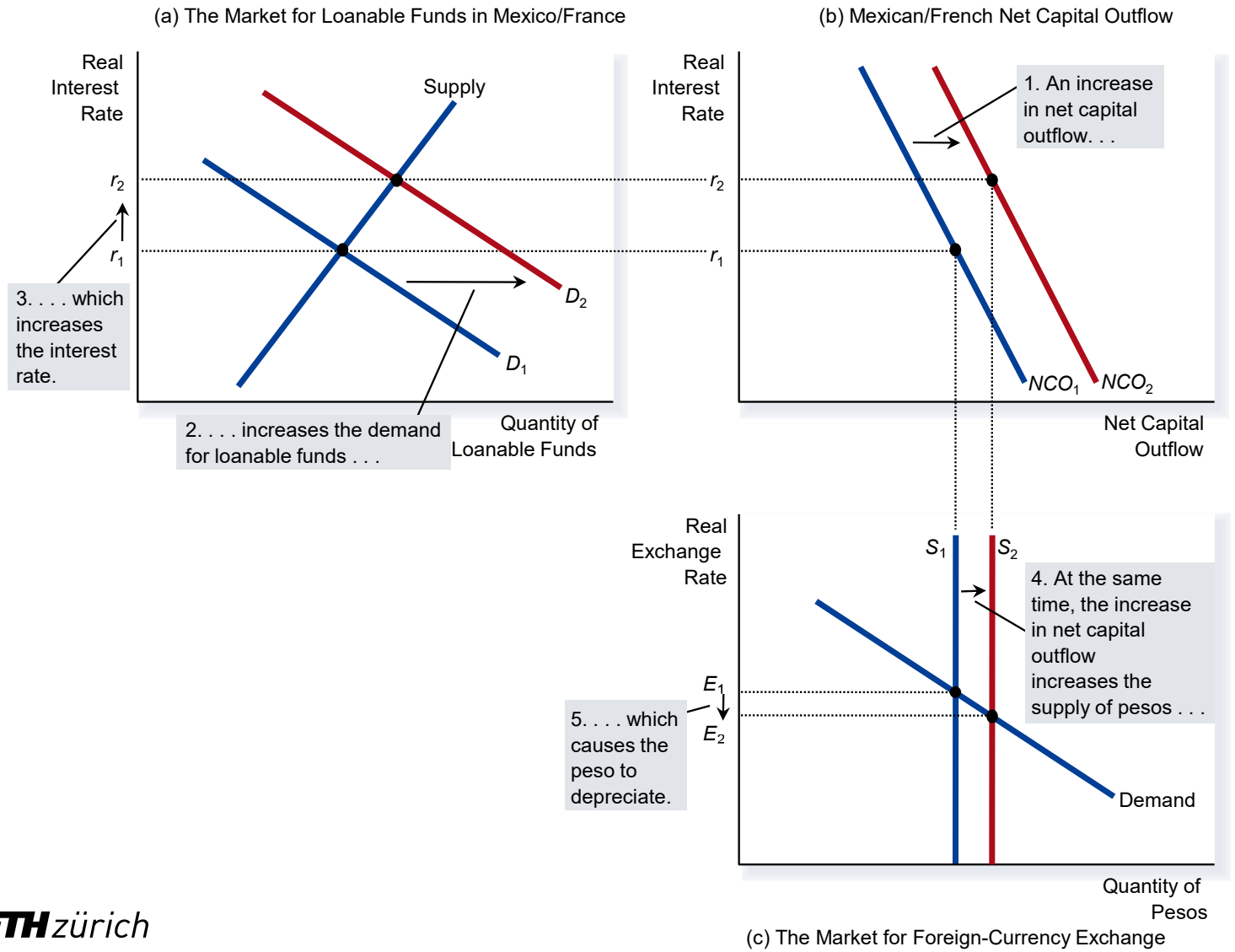
- Effect of an Import Quota
  - Trade policies do not affect the trade balance



Political Instability and Capital Flight

- ***Capital flight*** is a large and sudden reduction in the demand for assets located in a country
  - If investors become concerned about the safety of their investments, capital can quickly leave an economy
  - Interest rates increase and the domestic currency depreciates
- Capital flight has its largest impact on the country from which the capital is fleeing, but it also affects other countries
- When investors around the world observed political problems in Mexico in 1994, they sold some of their Mexican assets and used the proceeds to buy assets of other countries
  - This increased Mexican net capital outflow
    - The demand in the loanable funds market increased, which increased the interest rate
    - This increased the supply of pesos/euros in the foreign-currency exchange market, causing a depreciation

The Effects of Capital Flight



Summary

- Net exports are the value of domestic goods and services sold abroad minus the value of foreign goods and services sold domestically
- Net capital outflow is the acquisition of foreign assets by domestic residents minus the acquisition of domestic assets by foreigners
- An economy's net capital outflow always equals its net exports
- An economy's saving can be used to either finance investment at home or to buy assets abroad
- The nominal exchange rate is the relative price of the currency of two countries
- The real exchange rate is the relative price of the goods and services of two countries
- When the nominal exchange rate changes so that each Swiss Franc buys more foreign currency, the Swiss Franc is said to appreciate or strengthen
- When the nominal exchange rate changes so that each Swiss Franc buys less foreign currency, the Swiss Franc is said to depreciate or weaken

Summary

- According to the theory of purchasing-power parity, a unit of currency should buy the same quantity of goods in all countries
- The nominal exchange rate between the currencies of two countries should reflect the countries' price levels in those countries
- To analyze the macroeconomics of open economies, two markets are central—the market for loanable funds and the market for foreign-currency exchange
- In the market for loanable funds, the interest rate adjusts to balance supply for loanable funds (from national saving) and demand for loanable funds (from domestic investment and net capital outflow)
- In the market for foreign-currency exchange, the real exchange rate adjusts to balance the supply of Swiss Francs (for net capital outflow) and the demand for Swiss Francs (for net exports)
- Net capital outflow is the variable that connects the two markets
- A policy that reduces national saving, such as a government budget deficit, reduces the supply of loanable funds and drives up the interest rate
- The higher interest rate reduces net capital outflow, reducing the supply of Swiss Francs
- The Swiss Franc appreciates, and net exports fall

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Principles of Macroeconomics

Lecture 9: Business cycles

Prof. Dr. Jan-Egbert Sturm

Three Key Facts About Economic Fluctuations

1. Economic fluctuations are irregular and difficult to predictable
   - Fluctuations in the economy are often called the business cycle
   - These fluctuations do not follow regular or easily predictable patterns

2. Most macroeconomic variables fluctuate together
   - Most macroeconomic variables that measure some type of income or production fluctuate closely together
   - Although many macroeconomic variables fluctuate together, they fluctuate by different amounts

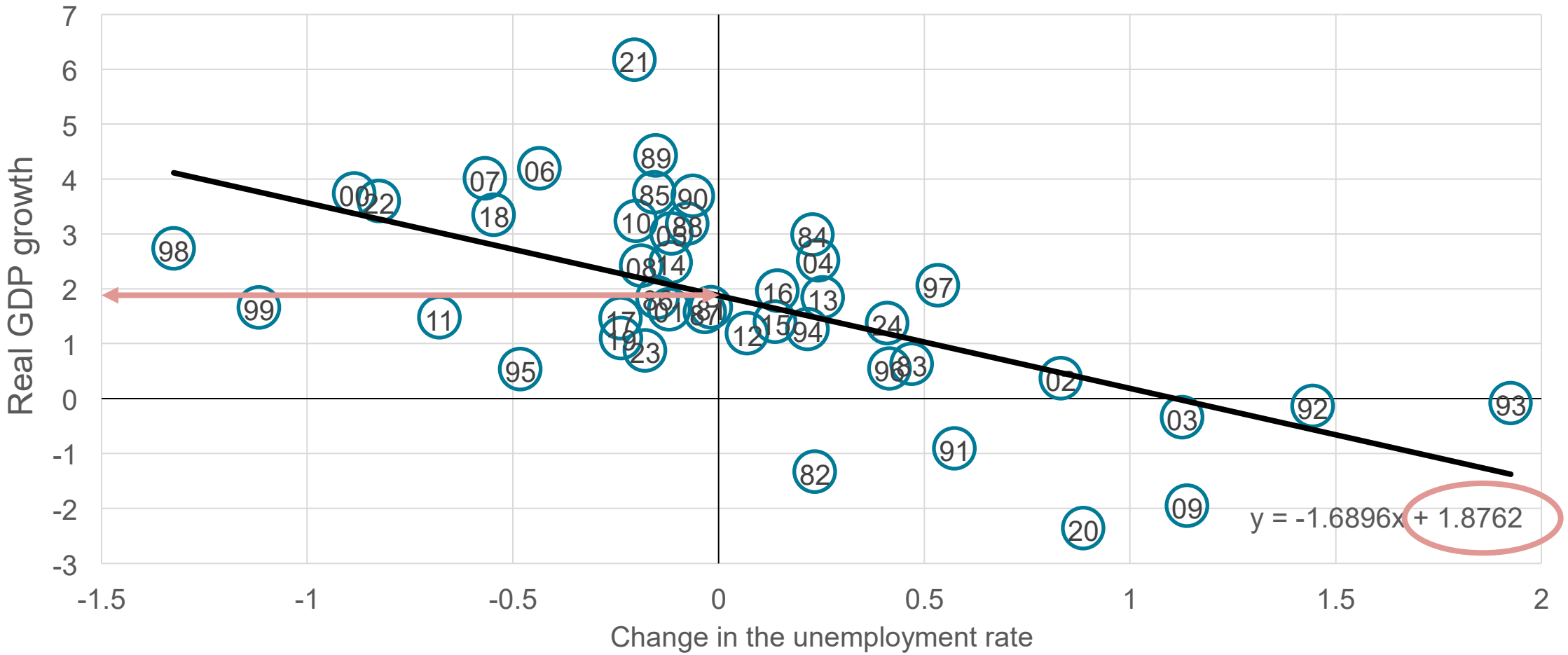
3. As output falls, unemployment rises
   - Changes in real GDP are inversely related to changes in the unemployment rate
   - During times of recession, unemployment rises substantially

Okun's Law

- ***Okun's law*** states that in order to keep the unemployment rate steady, real GDP needs to grow at or close to its potential

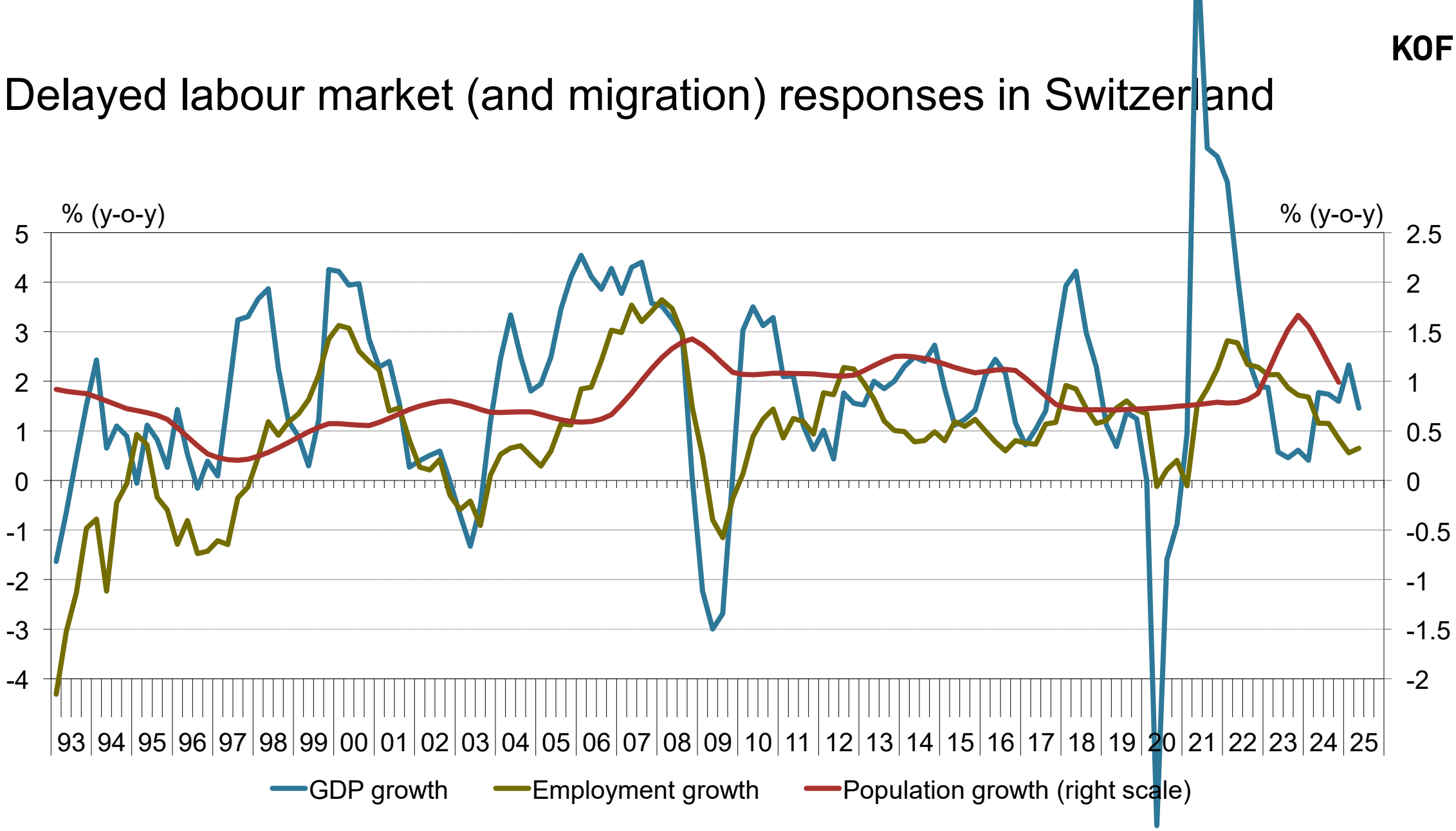
- There is a time-lag between any downturn in economic activity and a rise in unemployment and vice versa
  - Unemployment is a therefore a lagged indicator

Okun's Law for Switzerland (1981-2024)



Sources: Seco, SFSO

Delayed labour market (and migration) responses in Switzerland

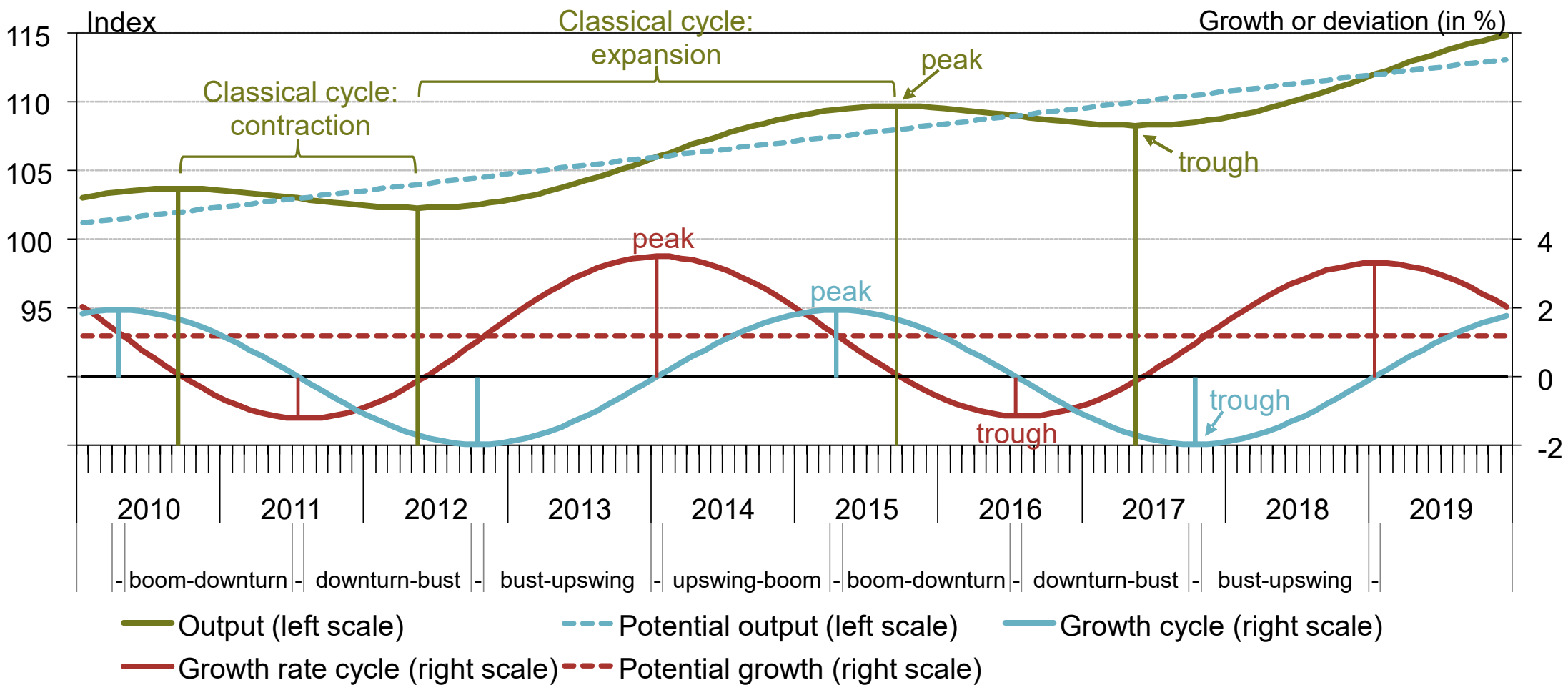


Sources: Seco, SFSO, KOF

Economic Activity Fluctuates from Year to Year

- The level of economic activity, the amount of buying and selling, changes over time
- ***Recession*** is a period of declining real incomes and rising unemployment.
  - Often defined as a period with at least two successive quarters of negative economic growth
- ***Depression*** is a severe recession
- Central to the analysis of business cycles is GDP over time
  - GDP shows a pattern on peaks and troughs and periods where growth is accelerating, decelerating and in some cases is declining
    - A peak is where economic activity reaches a high and real output begins to decline
    - A trough is where economic activity reaches a low and the decline ends
- In practise there are three definitions of the ***business cycle*** that are widely (and interchangeably) used
  - Classical cycle: Looking at the level of GDP
  - Growth cycle: Looking at the deviation of GDP from potential (/trend)
  - Growth rate cycle: Looking at the growth of GDP

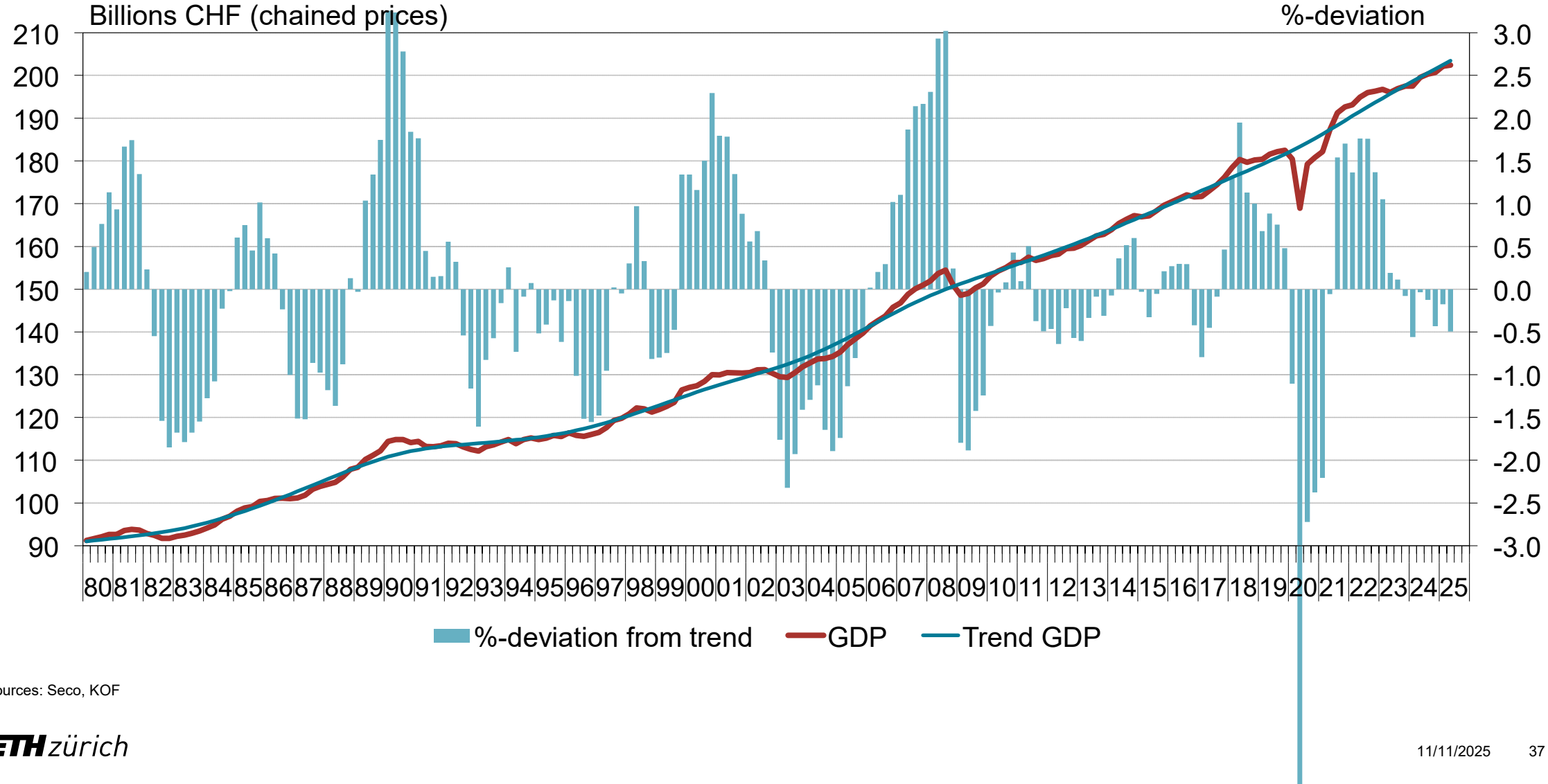
Different measures/concepts of the business cycle



Business cycle phases when using the Growth Cycle concept

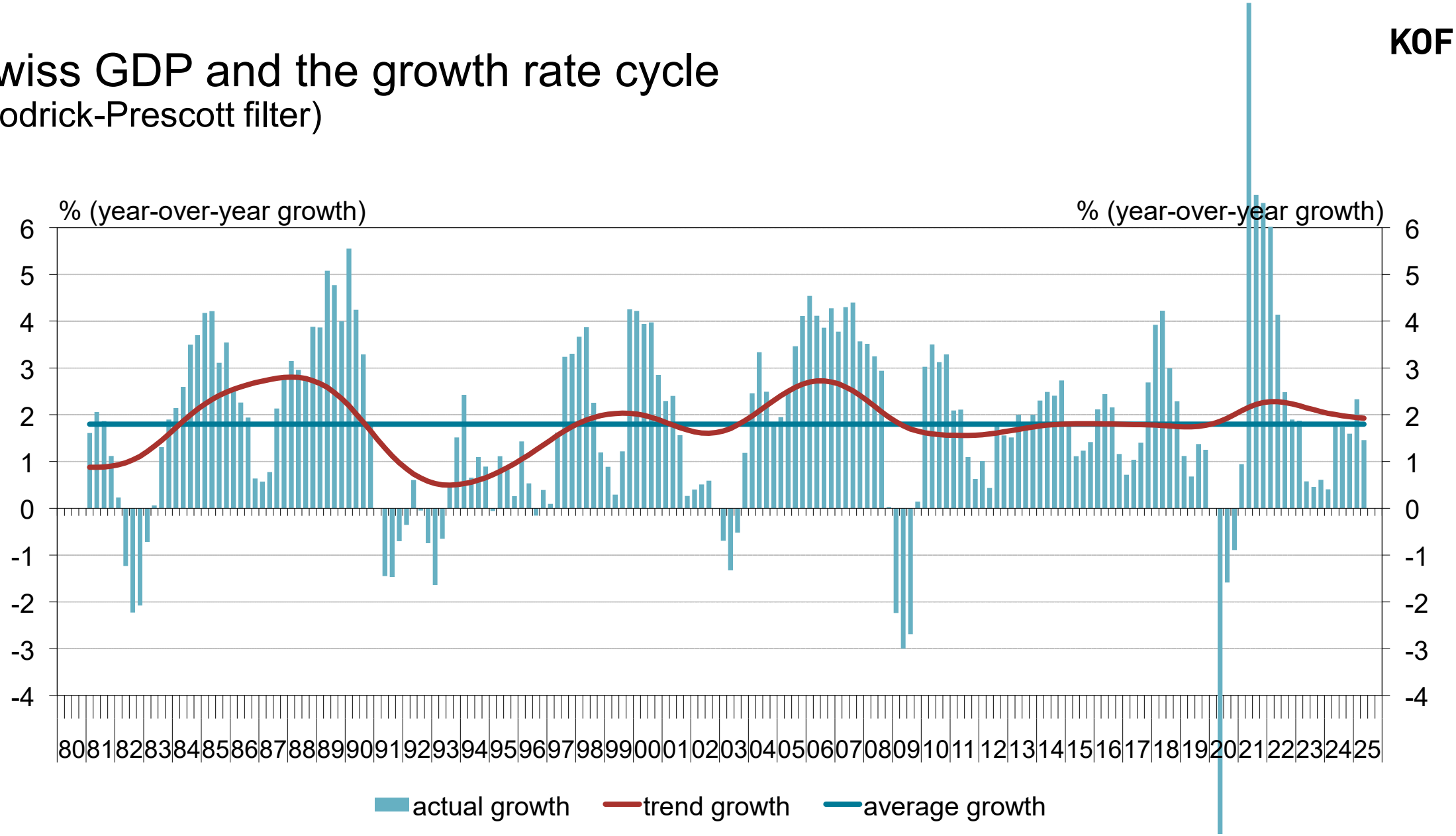
	Bust - upswing	Upswing - boom	Boom - downturn	Downturn - bust
Economic activity	• Gross Domestic Product (GDP), industrial production, and other measures of economic activity levels are below potential, while growth is already above normal.	• Activity measures show above-average, albeit declining, growth rates. Production levels are higher than normal.	• Activity measures show further decelerating and now below average rates of growth. Production levels are still higher than normal.	• Activity measures are below potential. Albeit growth is lower than normal, it is picking up again.
Capital spending	• Some recovery visible in investment goods. Upturn most pronounced in orders for light producer equipment.	• Capital spending expands rapidly.	• The growth of capital spending slows down.	• Decreased spending most evident in orders for new business equipment.
Employment	• Layoffs are happening, but at a reduced pace and new hiring does not yet occur. The unemployment rate is rising	• Business turn to overtime and temporary employees. Some rehiring as overtime hours rise. The unemployment rate stabilises and starts falling	• Business slows its rate of hiring. The unemployment rate continues to fall but at slowly decreasing rates.	• Business first cuts hours and freezes hiring, followed by outright layoffs. The unemployment rate starts to rise.
Inflation	• Inflation continues to fall.	• Inflation picks up modestly.	• Inflation further accelerates.	• Inflation decelerates but with a lag.

Swiss GDP (in levels) and the business cycle – using the Growth Cycle concept (and the Hodrick-Prescott filter)



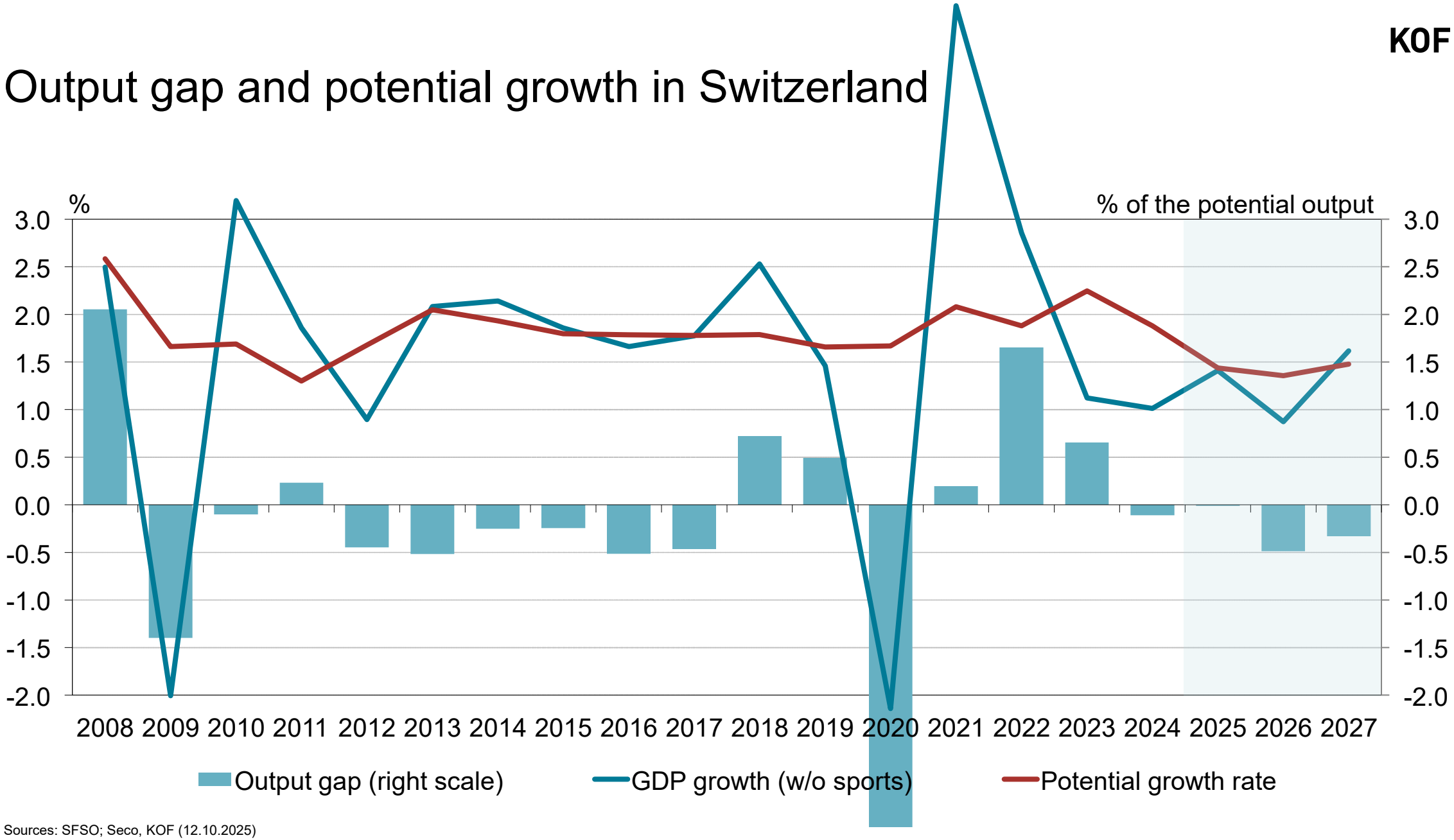
Sources: Seco, KOF

Swiss GDP and the growth rate cycle
(Hodrick-Prescott filter)



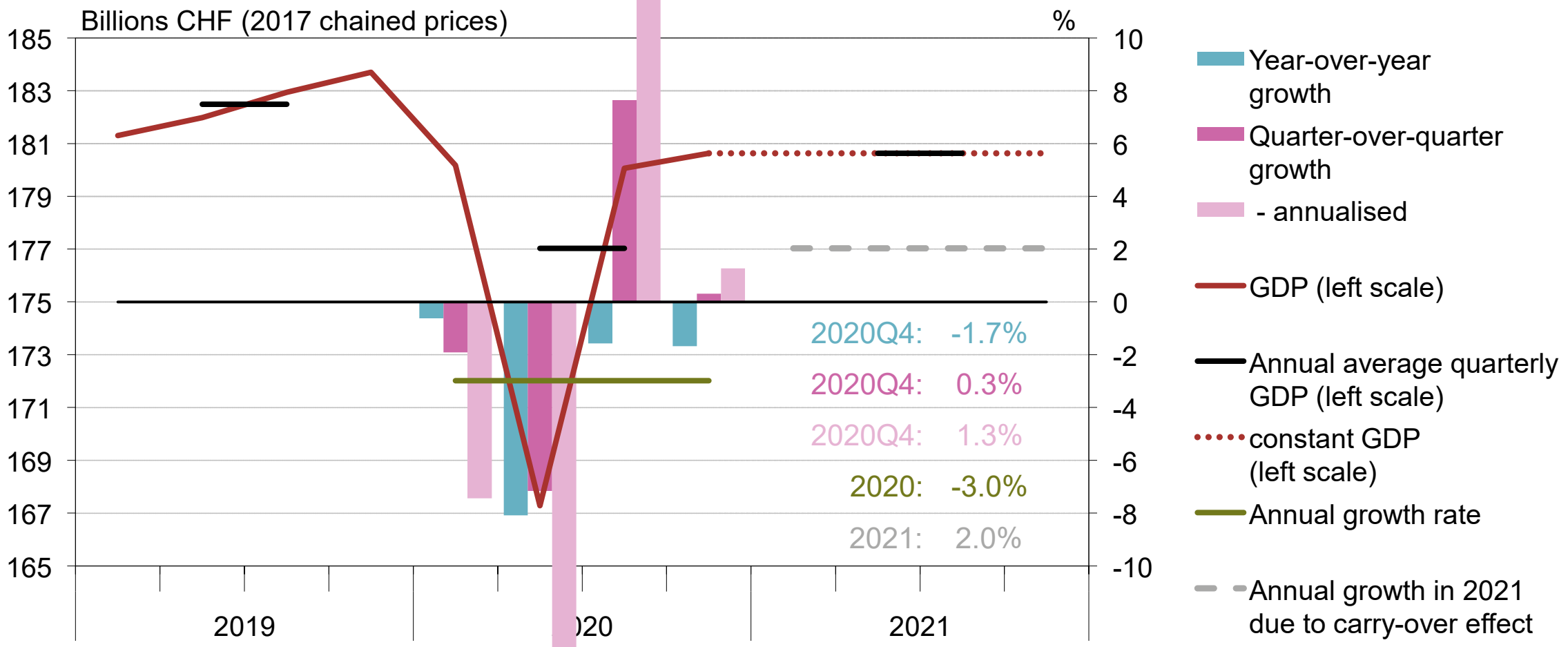
Sources: Seco, KOF

Output gap and potential growth in Switzerland



Sources: SFSO; Seco, KOF (12.10.2025)

Different growth rates at the end of 2020 in practice



Sources: Seco, KOF (data as of July 2021)

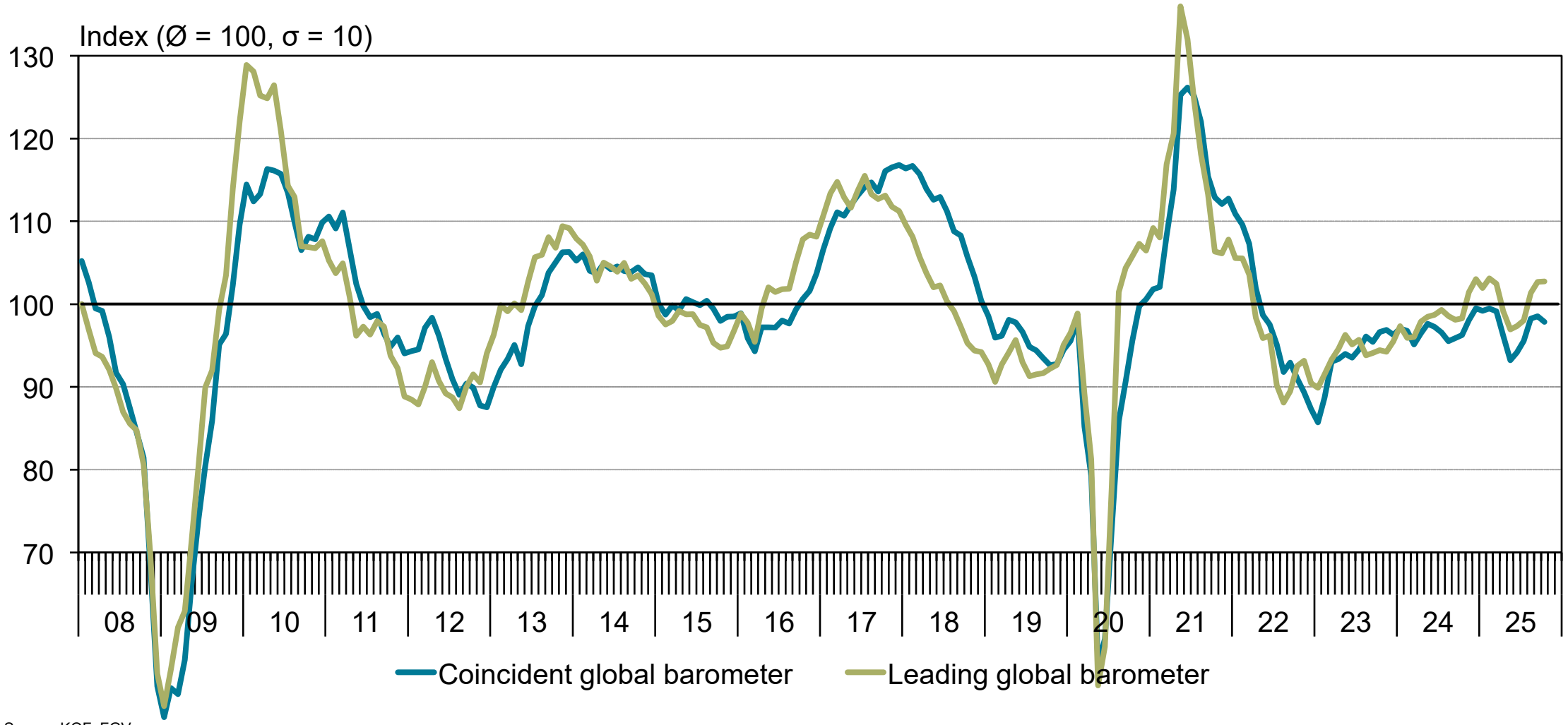
Trends

- There is some disagreement between economists about economic time series
- One such disagreement is about the nature of trends
- A ***trend*** is the underlying long-term movement in a data series
  - Most economic time series show an upward trend.
- ***Stationary data*** is time series data that has a constant mean value over time
  - Deterministic trends are independent of time for the series being analysed
- ***Nonstationary data*** is time series data where the mean value can either rise or fall over time
  - Stochastic trend variables change by some random amount in each time period
- Assuming GDP shows a deterministic trend, rising 1¾% on average in Switzerland, then temporary policy measures can be applied when it deviates from trend
- If the data exhibits a trend that is stochastic then temporary policy measures might be applied that are unnecessary and possibly distorting

Procyclical and Countercyclical Movements in Macroeconomic Data

- ***Comovement*** refers to the movement of pairs of variables.
- Economists compare another economic variable such as inflation or employment with GDP over time and see if any relationship can be determined
- ***Procyclical*** is a variable that is above trend when GDP is above trend.
  - Real wages is an example – it tends to increase faster during booms
- ***Countercyclical*** is a variable is that is below trend when GDP is above trend.
  - Unemployment is an example since unemployment tends to fall as GDP grows
- Economicsts use cyclical indicators to identify potential turning points in economic activity
  - A ***coincident indicator*** occurs at the same time as changes in economic activity
  - A ***leading indicator*** can be used to foretell future changes in economic activity
  - A ***lagging indicator*** occurs after changes in economic activity have occurred

The KOF-FGV Global Economic Barometers



Source: KOF, FGV

Causes of Changes in the Business Cycle

1. Household spending decisions
2. Firms' decision making
3. External sources
4. Government policy
5. Confidence and expectations

1. Household Spending Decisions

- Households decide on how much labour to supply
  - The amount of labour supplied depends in part on the real wage rate
- The rate of growth of wages in relation to prices affects consumers' purchasing power
- Households will also make decisions based on changes in interest rates, house prices and taxation
  - Increases in interest rates may encourage a rise in saving and a reduction in consumption
  - People borrow on the strength of the value of their property
  - Tax rates changes affect different people in different ways

2. Firms' Decision Making

- If firms face strong demand they are likely to:
  - increase output
  - take on more workers
  - Buy more raw materials and semi-finished goods
- Some firms may invest in new equipment and new premises or even acquire other firms
- If the real wage rate falls then firms can afford to hire more workers
- If productivity levels rise then costs can be lowered and firms can be more competitive
- Firms will respond to changes in inventory levels by expanding or contracting output as necessary

3. External Sources

- Movements in exchange rates affect the competitiveness of domestic and foreign firms through changes in import and export prices
- Economic activity abroad can impact on countries as consumption and investment decisions by firms and consumers abroad change
- Unpredictable events such as war and natural hazards can have a big impact

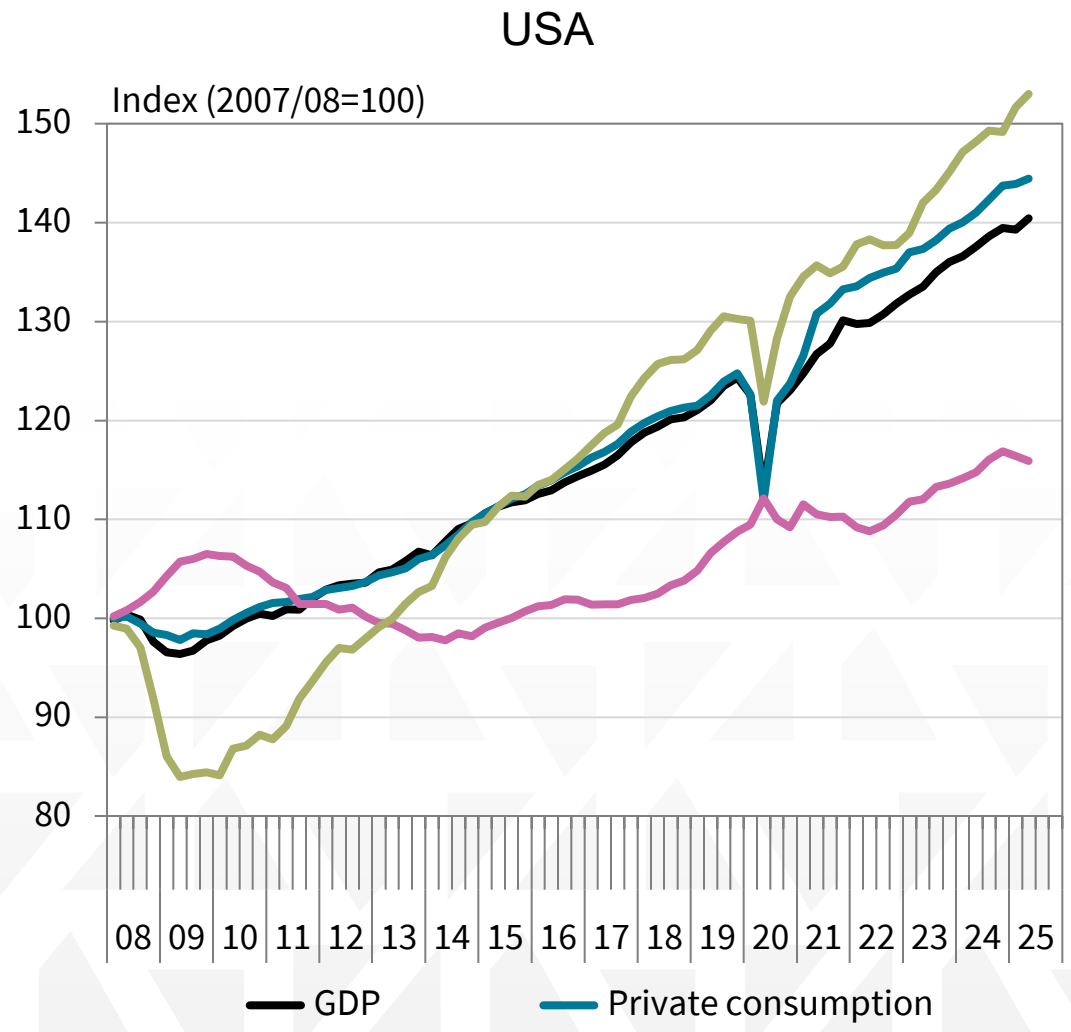
4. Government Policy

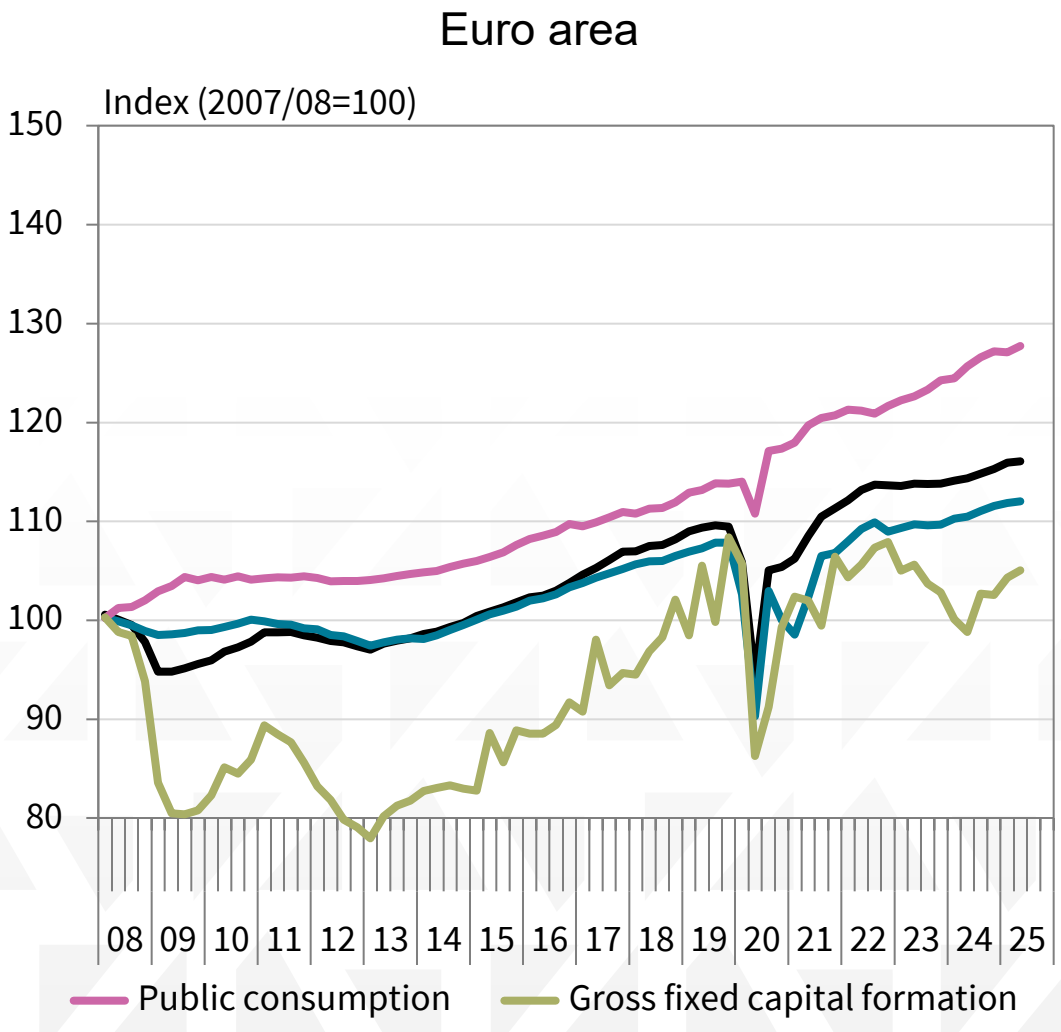
- Governments collect taxes and spend the revenues
- Changes in taxation and government spending (including infrastructure spending) affect the economy
- Although independent, central banks have targets to achieve
  - Changes in interest rates will affect both households and firms and policy decisions like quantitative easing and measures designed to help ease credit flows to business and households

5. Confidence and Expectations

- Households and firms not only make decisions based on current needs but also on the future
- Expectations and confidence of the future shape decisions by firms and households and can play a significant part in swings in economic activity

Comparing the US and the euro area: a demand-side view





Source: BEA, Eurostat

Business Cycle Models

1. Labour market side – the (new) classical models
2. Labour market side – the (new) Keynesian models
3. Product market side – the (new) classical models
4. Product market side – the (new) Keynesian models
5. Supply side – Real Business Cycle models
   - Production factors determine supply on goods and service markets

- If markets are assumed to clear: (new) classical models (or RBC models)
- If markets are not assumed to clear: (new) Keynesian models

1. Labour market side – the (new) classical models

- Labour markets clear but workers have ***imperfect information***
  - Based around the concept of ***anticipated*** and ***unanticipated*** price changes
- If price rises are ***anticipated*** (i.e. ideal world without this form of business cycle)
  - Workers will recognize that real wages would fall without a nominal wage increase and will supply less labour
  - The demand for labour will be greater than the supply of labour, so nominal wages will rise
  - This will keep the real wage effectively constant at the new equilibrium
- If workers ***do not anticipate*** (and do not realise) the change in prices
  - Real wages fall, firms demand more labour but workers continue to offer the same amount of labour
  - The demand for labour will be in excess of supply and nominal wages will rise but less than prices
  - The result is that output will rise but the real wage will fall (real wages as countercyclical)

2. Labour market side – the (new) Keynesian models

- Keynes believed markets do not clear quickly
  - Markets are impaired by the existence of (sticky prices and) ***sticky wages***
- Nominal wages are often slow to adjust in the economy due to long-term employment contracts
  - An increase in prices will make real wages lower, increasing labour demand and output
    - Real wages are countercyclical

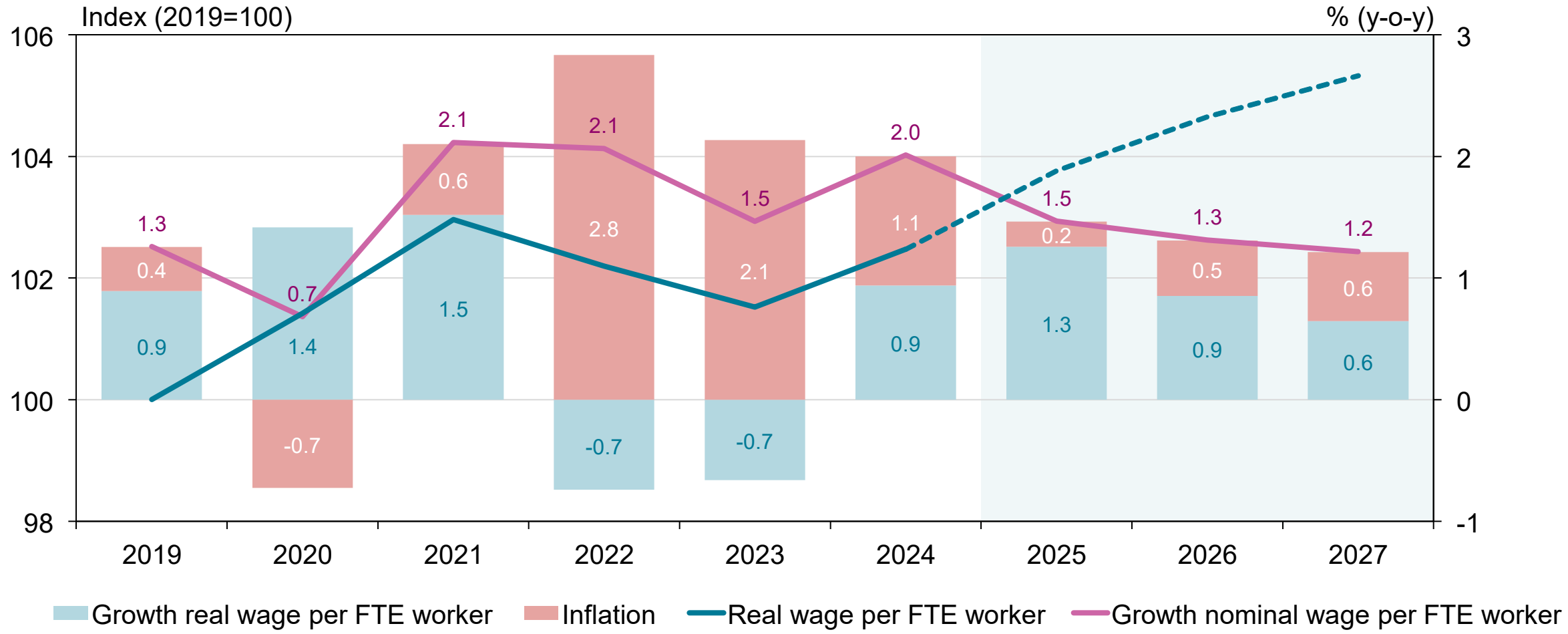
3. Product market side – the (new) classical models

- Information problems might lead to sub-optimal decisions
- Firms might not be able to distinguish between inflation and a relative price change
- An observed increase in the price of the goods they supply might lead them to produce more

New Classical models are driven by asymmetric information

- The New Classical interpretation rests on this ***inflation fallacy***
  - Workers misinterpret a rise in nominal wages as a rise in real wages
  - Firms misinterpret a rise in prices as an increase in demand for their goods
- The economy moves to a temporary equilibrium where the expectations of some economic actors are not fully incorporated because they are incorrect
  - Over time, workers realize that real wages have changed and negotiate for wage rises
  - Firms' costs rise and some firms cut back supply
  - The economy returns to trend output but at a higher price level once expectations have fully adjusted
- Cyclical implications when output is above trend:
  - Unemployment is countercyclical and employment will be above trend and so be procyclical.
  - In case of asymmetric information problems on the labour market, inflation is procyclical and real wages are countercyclical because as output rises real wages fall
  - If there is an information problem on the goods/service market, then real wages move procyclical. The boom creates higher labour demand. Without information problem on the labour market, this implies that this higher demand will lead to the equilibrium real wage to be higher

Wage development in Switzerland – a short-run perspective



Quellen: SECO, KOF (12.10.2025)

4. Product market side – the (new) Keynesian models

- The prices of some goods and services can also be slow to respond to changes in the economy
  - This can be due to menu costs or contractual arrangements
  - Because not all prices adjust instantly to changing conditions,
    an unexpected increase in the price level leaves some firms with lower-than-desired prices
  - This increases sales and cause firms to increase the quantity of goods and services supplied
- In the short run, output increases above trend but prices do not change because of being sticky
- Over time, the economy returns to trend, because firms will eventually be able to raise prices

- Employment, real wages and inflation are procyclical and unemployment is countercyclical

5. Supply side – Real Business Cycle models

- In this model both ***positive and negative technology shocks*** affect productivity
- Assume there are no market imperfections and firms and households are profit and utility maximizing
- A negative technology shock would reduce labour productivity and the demand for labour fall
  - Output also drops and unemployment increases
  - Output falls because aggregate supply falls
  - Prices rise
- When aggregate supply shifts, the expectations of economic actors are still correct
  - (Effectively it is the long-run supply curve that shifts)
- Therefore there is no reason for governments or central banks to intervene
- In real business cycle models, employment, labour productivity and real wages are procyclical

Summary

- Economies experience periods of changing levels of economic activity
- Key macroeconomic data is often time-series data which raises questions about the validity and reliability of statistical tests applied to data sets
- Economic growth appears to follow a trend which rises over a period of time
- Deviations from this trend are known as the business cycle
- The business cycle has characteristic features such as peaks in economic activity, slowdown, troughs and upturns with key turning points
- Trends can be stationary and nonstationary. Nonstationary, data can exhibit deterministic trends which change by a constant amount independent of time or be stochastic where the trend variable changes by a random amount
- Macroeconomic data can be viewed in pairs with one of the pairs being GDP
  – The variable compared may be procyclical or countercyclical
- Collecting macroeconomic data allows economists to view indicators as leading, coincident or lagging

Summary

- Changes in the business cycle can be caused by changes in household and firms' decision making, by external shocks, government policy and changes in confidence or expectations of the future
- Business cycle models differ in the assumptions they make about the extent to which markets clear and the relationship between the supply side and demand side of the economy
- Real business cycles emphasize the effects of changes in technology as causes of changes in economic activity